Variable Acceleration

Questions 1–29

Question 1. A particle moves along a straight line so that its displacement s metres at time t seconds, where $t \geq 0$, is given by each of the following functions. Find expressions for the velocity and the acceleration of the object at time t for:

a) $s = 4t^3 - 5t$
b) $s = t^5 + 2t^2$
c) $s = t(t-1)$
d) $s = 2t - \frac{1}{t}$

Question 2. A particle starts at rest before moving along a straight line. Its velocity v ms^{-1} at time t seconds is given by each of the following functions, where $t \geq 0$. In each case find the displacement in terms of t:

a) $v = 8t^3 - 6t$
b) $v = 6t^2 + 2t^3$
c) $v = \frac{1}{10}t^2(t-5)$
d) $v = t - \frac{1}{t^2}$

Question 3. A particle is moving along a straight line. Its acceleration a ms^{-2} at time t seconds is given by each of the following functions, where $t \geq 0$.
In each case find the velocity at t in terms of t, given that is known that the particle is moving with velocity v_1 at time t_1

a) $a = 15t^4 - 4t, t_1 = 1, v_1 = 2$
b) $a = 9t^2 + 5t^4, t_1 = 2, v_1 = 40$
c) $a = -0.09t(t - 30), t_1 = 1, v_1 = 3$
d) $a = \sqrt{t} - \frac{1}{t^3}, t_1 = 1, v_1 = 2$

Question 4. A car drives between two sets of traffic lights, stopping at both. Its speed v ms^{-1} at time t s is modelled by

$$v = -\frac{1}{50}t^2 + \frac{6}{5}t, \qquad 0 \leq t \leq 60$$

a) Find the times at which the car is stationary.
b) Sketch the velocity-time graph for the motion of the car.
c) Find the distance between the two sets of traffic lights.
d) Find an expression for acceleration a ms^{-2} in terms of t
e) Explain why the model is no longer appropriate after 60 seconds.

Question 5. A particle's acceleration a ms^{-2} with time t seconds is given by

$$a = 15\sqrt{t} + 2, \qquad t \geq 0$$

The initial velocity is 1 ms^{-1} and when $t = 1$ the displacement of the particle is 6 m.

a) Find an expression for the velocity v with time t

b) Find an expression for the displacement s with time t

Question 6. A particle's acceleration a ms^{-1} at any given time t seconds is given by

$$a = -30t^2 + 25t^3, \qquad t \geq 0$$

The particle initially starts at position O, with a velocity of 5 ms^{-1}.

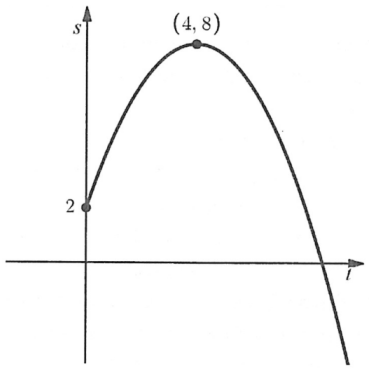
a) Find an expression for the velocity given the particle initially starts at position O.

b) Find the displacement of the particle when 2 seconds have passed.

Question 7. The displacement s (metres) over time t (seconds) for a particle is given by

$$s = -\frac{3}{8}t^2 + kt + 2, \qquad t \geq 0$$

where k is a positive constant and is shown in the diagram below.



a) Show that $k = 3$
b) State the distance travelled by the particle when $t = 4$
c) Find the time at which the particle returns to its initial position.
d) Find an expression for the velocity v in terms of t
e) Find the speed of the particle when $t = 10$ seconds.

Question 8. Prerequisite knowledge for Q8: $F_{\text{net}} = ma$ An object of mass 0.4 kg is moving in a straight line under the action of a resultant force F N. Its velocity v ms^{-1} at any given time t seconds may be given by

$$v = 8t - 3t^2, \qquad t \geq 0$$

a) Find an expression for the object's displacement, s metres, in terms of t
b) Find an expression for the object's acceleration, a ms^{-2}, in terms of t
c) Find an expression for F in terms of t

Question 9. The velocity, v ms^{-1} at any given time t seconds of a car may be given by

$$v = kt - 3 + \frac{1}{t^2}, \qquad \frac{1}{2} \leq t \leq 10$$

where k is an unknown positive constant.
It is known that the acceleration when $t = 2$ is 1.75 ms^{-2}.

a) Find the value of k
b) Find the total distance travelled by the car for $\frac{1}{2} \leq t \leq 10$

Question 10. Ellis is going to the bus stop. His displacement, s metres at any given time t seconds is

$$s = -\frac{1}{k}t(t - 240), \qquad 0 < t \leq 120$$

where k is an unknown positive constant.
It is known that the velocity when $t = 20$ is 4 ms^{-1}.

a) Show that $k = 50$
b) Find the speed at which Ellis is travelling when $t = 60$ seconds
c) Find the speed at which Ellis is travelling when $t = 120$ seconds.
d) Explain in context what happens to Ellis' velocity and displacement if this same model is used for $t \geq 120$

Question 11. A particle's velocity, v ms^{-1} with time t seconds is given by

$$v = -t + 10, \qquad t \geq 0$$

Find when the particle will return to its original position.

Question 12. An object moves along a straight line so that its acceleration a ms^{-2} at time t seconds is given by

$$a = 4 + 3t - 6t^2, \qquad t \geq 0$$

a) Given that the object is initially stationary, find the velocity, v, in terms of t
b) Given that when 1 second has passed, the object has moved in a positive direction of 6 metres from its initial position, find the displacement, s in, in terms of t
c) Hence, find the time at which the object returns to its original position, giving your answer to 3 significant figures.

Question 13. A particle's displacement, s metres with time t seconds is given by

$$s = t^3 - 6t^2 + 12t, \qquad 0 \leq t \leq 10$$

a) Find the distance travelled by the particle over the 10 seconds.
b) Find the time when the particle is at rest.
c) Show that, according to this model, the particle will never reverse direction.
d) Show that it does not have a constant acceleration.

Question 14. A particle's acceleration, a ms^{-2} with time t seconds is given by

$$a = -1.8t + 18$$

where $t \geq 0$
Find the time it takes for the particle to travel 600 metres given it starts from rest.

Question 15. A car is being driven on a straight flat road, with velocity v ms^{-1} at time t seconds is given by

$$v = 2t - 0.001t^3, \qquad 0 \leq t \leq 25$$

a) Find the distance travelled by the car in the first 20 seconds.
b) Show that for $0 \leq t \leq 25$, the car is always accelerating (and not decelerating).

Question 16. The displacement, s metres at any given time t seconds of a car may be given by

$$s = \frac{4}{15}t^2 - \frac{1}{675}t^3, \qquad 0 \leq t \leq 120$$

a) Find the velocity of the car v in terms of t
b) Find the maximum speed for $0 \leq t \leq 120$

Question 17. Prerequisite knowledge for Q17-Q18: Constant Acceleration A car is being tested on a straight, flat road. Its velocity is found to be 12.5 ms^{-1} at 5 seconds and 25 ms^{-1} at 10 seconds. A model for its velocity v ms^{-1} against time t seconds is

$$v = at^2 + bt^3, \qquad t \geq 0$$

where a and b are unknown constants.

a) Find the values of a and b

A second model suggests that the cars motion may be modelled with constant acceleration.

b) Show that the distance travelled for both these two models in the first 10 seconds is the same.

Question 18. A runner having started from rest reaches a velocity of 3.5 ms^{-1} in 7 seconds.

a) Assuming constant acceleration, find a model for the velocity v ms^{-1} at any given time t

b) Find the distance travelled by the runner in the 7 second duration using this model.

An alternative model is given by

$$v = -\frac{1}{k}(t-7)^2 + 3.5, \qquad t \geq 0$$

where k is a positive constant.

c) Find the value of k

d) Find the distance travelled by the runner in the 7 second duration according to this second model.

e) By first sketching both models on a graph, explain why the second model gives a larger distance than the first.

Question 19. The velocity, v ms^{-1} at any given time t seconds of a cyclist for a 6 second section of their journey may be given by

$$v = \frac{2}{5}t^3 - 4t^2 + 10t, \qquad 0 \leq t \leq 6$$

a) Find the distance the cyclist has travelled in the first 5 seconds.
b) Find the maximum speed of the cyclist and the time at which it occurs.
c) Find the minimum acceleration of the cyclist and state what its value represents in context of the cyclist.
d) Explain why this model would not be valid in the long term.

Question 20. The velocity, v ms^{-1} at any given time t seconds of a person walking along a flat, straight path may be given by

$$v = \frac{k}{\sqrt{t}} + \frac{t}{80}, \qquad 0 < t \leq 120$$

where k is an unknown positive constant.
It is known that the acceleration when $t = 25$ is 0.0065 ms^{-2}.

a) Show that $k = \frac{3}{2}$
b) Find the total distance travelled by the person for $60 < t \leq 120$
c) Comment on the validity of the model for small values of t

Question 21. The velocity, v ms^{-1} at any given time t seconds of a particle may be given by

$$v = pt^2 + qt + r, \qquad t \geq 0$$

where p, q and r are unknown constants.
Initially, it has a velocity $v = 7.8$ ms^{-1}. It reaches its minimum velocity of 3 ms^{-1} when $t = 4$
Find the values of p, q and r

Question 22. The acceleration, a ms^{-2} at any given time t seconds of a particle may be given by

$$a = kt - 4, \qquad t \geq 0$$

where k is an unknown constant. The initial velocity is twice the magnitude of the minimum velocity which occurs when $t = 2$
Find the value of k and hence an expression for the velocity, v in terms of t only.

Question 23. The velocity, v ms^{-1} at any given time, t seconds, of a particle may be given by

$$v = \frac{3}{2}t^2 - \frac{9}{2}t + 3, \qquad t \geq 0$$

The particle is initially at the origin.
Determine the displacement of the particle from the origin when it has covered a total distance of 8.5 metres.

Question 24. The velocity v ms^{-1} at any given time t seconds of a particle starting at initial position O may be given by

$$v = -t^2 + kt, \qquad 0 \leq t \leq 9$$

where k is an unknown positive constant.
Initially, the acceleration is 6 ms^{-2}.

a) Show that $k = 6$
b) Show that the maximum velocity occurs when $t = 3$
c) Find the distance it has travelled when it has reached its maximum velocity.
d) Find the displacement of the particle over the 9 second duration.
e) Find the time at which the particle is not moving.
f) Find the distance travelled over the 9 second duration.

Question 25. The velocity v ms^{-1} at any given time t seconds of a cyclist may be given by

$$v = \begin{cases} 0.81t^2 - 0.09t^3 & 0 \leq t \leq 6 \\ 9.72 & 6 \leq t \leq 20 \end{cases}$$

a) State the initial velocity.
b) Describe the motion of the cyclist.
c) Find the maximum acceleration of the cyclist.

Question 26. Filip is walking to a shop which is 800 m from his house, but on arriving he realises he left his phone at home and returns to retrieve it. His displacement from his house s m at any given time t seconds may be given by

$$s = \begin{cases} \frac{3}{200}t^2 - \frac{1}{40000}t^3 & 0 \leq t \leq t_1 \\ -3.2t + 2080 & t_1 \leq t \leq t_2 \end{cases}$$

where t_1 is the time at which Filip arrives at the shop, and t_2 is the time at which Filip arrives home. According to the model,

a) find the time t_1 it takes to reach the shop.
b) find the time t_2 taken for the whole journey, giving your answer to the nearest second.
c) find the maximum velocity during his journey to the shop.
d) find the maximum speed for his full journey.
e) Comment on the validity of the model for the return journey.

Question 27. Prerequisite knowledge in Q27-Q28: Differentiation of the form ae^{kx} A school is holding a 20 m race. The displacement s m at any given time t seconds of the fastest child running the race is given by

$$s = 6t - 15 + 15e^{-0.4t}, \qquad 0 \leq t \leq k$$

a) Show that the time k seconds according to the model, at which the fastest child finishes the 20 m is approximately 5.56 s.
b) Find an expression for the velocity v ms^{-1} in terms of t
c) Find the initial velocity of the child.
d) Find the maximum acceleration a ms^{-2} of the child during the race.

Question 28. The vertical displacement s metres above ground level of a carriage in "The Drop", a ride at an amusement park, at any given time t in the first 40 seconds may be given by

$$s = 0.8e^{0.1t} + 8e^{-0.2t}, \qquad 0 \leq t \leq 40$$

The ride stops at 40 seconds before dropping.
a) State the height above the ground at which the ride begins.
b) Find the time at which the ride changes direction, giving your answer to the nearest second.
c) Find the distance the ride drops before travelling upwards again.
d) Find an expression for the vertical acceleration a ms^{-2} in terms of t.
e) Find the maximum vertical speed of the ride according to the model.

In the second phase of the ride at $t = 40$, the ride drops in free-fall for 2 seconds before applying a brake.
f) Using $g = 9.8$ ms^{-2} determine the distance the carriage drops before the brake is applied.

Question 29. Prerequisite knowledge for Q29: $F_{\text{net}} = ma$ An object of mass 0.2 kg is moving in a straight line under the action of a resultant force F N. Its displacement s m at any given time t seconds may be given by

$$s = -\frac{1}{10}t^2(t-4), \qquad t \geq 0$$

Find an expression for F in terms of t